

Smart Lender: Project Planning & Lifecycle Framework

1. Project Workflow Flow Representation

The engineering roadmap follows a rigorous milestone-driven sequence. Each phase transitions inputs directly into downstream pipeline processes as mapped below:

[EPIC 1: Data Architecture Design] | ▼ (Sourced Data Matrices & Schema Configuration) [EPIC 2: Visualizing & Analyzing Data (EDA)] | ▼ (Exploratory Value Correlations & Statistics Reports) [EPIC 3: High-Fidelity Data Pre-Processing] | ▼ (Imputed, Balanced, and Scaled Training/Testing Sets) [EPIC 4: Model Building & Benchmarking] | ▼ (Optimized Model Binaries & Accuracy Metrics Weights) [EPIC 5: Web UI Application & Cloud Release]

2. Milestone Epic & User Story Framework

Execution sequences are split across five primary core epics to ensure systematic implementation:

Epic 1: Data Collection and Architecture Design

- **Story 1:** Ingest target operational risk dataset configurations into local repository directories.
- **Story 2:** Map database integration interfaces establishing structural bonds between user profiles, credit scores, and inference tracking pools.

Epic 2: Visualizing and Analyzing the Data (EDA)

- **Story 1:** Apply data parsing structures to audit data types, identify duplicates, and manage structural outliers.
- **Story 2/3/4:** Generate univariate data spread logs, bivariate charts, and multivariate heatmaps to isolate critical dependencies.

Epic 3: Data Pre-Processing Pipeline

- **Story 1/2:** Implement zero-value imputation rules and resolve minority-class imbalance issues via synthetic sampling.
- **Story 3/4:** Enforce min-max mathematical scaling rules and allocate static indices to create distinct training and evaluation arrays.

Epic 4: Model Development & Performance Benchmarking

- **Story 1-4:** Execute parallel training iterations for Decision Tree, Random Forest, KNN, and XGBoost structures.
- **Story 5:** Compute confusion matrices and serialize the highest-performing model binary configuration ready for cloud deployment.

Epic 5: Application Development & Containerized Release

- **Story 1/2:** Construct responsive form layouts matching database attributes and bind forms to an asynchronous Flask model execution layer.
- **Story 3:** Configure Docker multi-layer image dependencies to compile a public, hot-reloadable playground environment on Hugging Face Spaces.